Policy-Hard, Value-Easy: Inverted Actor-Critic Asymmetry in Deep Reinforcement Learning Based Assistive Indoor Navigation

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Abstract

Indoor navigation for visually impaired individuals remains a critical unsolved challenge, as GPS-denied environments demand intelligent, infrastructure-free guidance systems. This paper presents a comparative analysis of two deep reinforcement learning paradigms (value-based Deep Q-Networks (DQN) and policy-gradient Proximal Policy Optimization (PPO)) applied to indoor obstacle avoidance and pathfinding. Using a custom partially observable grid-world environment simulating residential spaces, we evaluate these algorithms' learning navigation policies from resource-constrained sensory inputs. A central contribution is the identification and validation of an *inverted asymmetry* architecture in actor-critic methods. Contrary to conventional heuristics prioritizing critic capacity, we demonstrate that a deeper policy network coupled with a streamlined value network yields superior performance in tasks with sparse rewards and complex geometric constraints. The optimized PPO agent achieved an average reward of 41.1 and reduced path traversal steps by approximately 50% compared to the best-performing DQN agent (14.3 vs. 28.6 steps). While DQN demonstrated unmatched reliability with 100% success rate and zero collisions, its policies were markedly conservative. PPO exhibited exceptional path optimality and generalization. Our findings propose a new architectural standard for deployment on resource-constrained assistive devices, with implications for the broader reinforcement learning community.

1 Introduction

Navigation is a fundamental human activity, integral to independence and quality of life. For the estimated 2.2 billion people globally living with vision impairment, navigating unfamiliar indoor environments poses significant physical and psychological challenges [Feghali et al., 2024]. Unlike outdoor environments, where Global Navigation Satellite Systems (GNSS) provide reliable positioning, indoor spaces are GPS-denied, structurally complex, and populated with dynamic obstacles. Traditional mobility aids, such as the white cane and guide dogs, provide essential immediate feedback but lack the capability for global path planning, semantic environmental understanding, and proactive obstacle avoidance [Larsen et al., 2021]. Consequently, the visually impaired (VI) community faces a "mobility gap," often relying on sighted assistance or memorization to traverse complex public and private spaces.

The advent of autonomous systems and computer vision has spurred the development of Electronic Travel Aids (ETAs) designed to bridge this gap. Early iterations relied on simple range-finding sensors (ultrasonic or infrared) to provide haptic or auditory feedback regarding proximity to obstacles [Plikynas et al., 2020]. While beneficial, these systems often suffer from a "data overload" problem, where the user is inundated with raw sensor data requiring significant cognitive effort to interpret. The next generation of ETAs aims to shift this cognitive burden from the user to the device, necessitating an intelligent agent capable of perceiving the environment, making navigational decisions, and conveying high-level instructions (e.g., "turn right at the doorway") rather than raw distance metrics [Srinivasaiah et al., 2024]. One promising paradigm for building such intelligent agents is deep reinforcement learning (DRL), which enables an agent to learn optimal decision-making through direct interaction with its environment.

Reinforcement learning (RL) has emerged as a dominant paradigm for robotic control, particularly for tasks involving high-dimensional state spaces and complex dynamics. In the context of navigation, the problem is typically modeled as a Partially Observable Markov Decision Process (POMDP), where the agent must infer the state of the world from limited sensor data and select actions to maximize a cumulative reward signal [Zhu et al., 2025]. Two primary families of algorithms have shown promise. Value-Based Methods, exemplified by the Deep Q-Network (DQN) and its variants, estimate the action-value function $Q(s, a)$, predicting the expected future reward for taking a specific action in a given state. The policy is derived implicitly by selecting the action with the highest Q-value. DQN has demonstrated remarkable stability and sample efficiency in discrete action spaces [La Fuente Neil and Guerra, 2024]. Alternatively, policy-gradient methods such as Proximal Policy Optimiza-

tion (PPO) optimize the policy directly by ascending the gradient of the expected return. These methods, particularly in Actor-Critic configurations, are capable of learning stochastic policies advantageous in environments requiring robust exploration or continuous control [Tan, 2025]. While both approaches have been extensively benchmarked in gaming environments (e.g., Atari, MuJoCo), their comparative performance in safety-critical, resource-constrained assistive navigation tasks remains an area of active research. The trade-off between safety (collision avoidance) and efficiency (path optimality) is of paramount importance in assistive robotics, where a collision represents not just a lower score, but a potential injury to the user [Hsu et al., 2021].

This research documents the development, training, and evaluation of a DRL-based navigation agent designed for the specific constraints of visually impaired assistance. The primary objective is to determine which RL architecture (DQN or PPO) provides the optimal balance of safety, efficiency, and generalization when navigating a simulated indoor grid world. The specific contributions include:

- Development of a `gym`-compatible grid-world environment abstracting key topological features of indoor spaces into a format suitable for accelerated training.
- Identification of a non-standard Actor-Critic architecture for PPO, termed **inverted asymmetry**, where the Policy network (Actor) is significantly deeper than the Value network (Critic), challenging the prevailing wisdom that prioritizes Critic capacity.
- A rigorous empirical comparison of DQN and PPO revealing distinct behavioral profiles: DQN exhibits "safety-first" behavior with zero collisions but suboptimal pathing, while PPO achieves near-optimal efficiency.
- A detailed analysis of the impact of network depth, learning rates, and exploration schedules on agent convergence, providing tuned hyperparameters for reproducing these results in similar domains.

The findings suggest that for discrete, grid-based navigation tasks with sparse rewards and strict topological constraints, policy-gradient methods with high-capacity policy networks offer the most promising path toward human-level navigation efficiency.

2 Literature Review

2.1 Evolution of Indoor Navigation Technologies

The pursuit of reliable indoor navigation has traversed several technological epochs. Early systems, often termed "Sensory Substitution Devices" (SSDs), aimed to translate visual information into auditory or tactile stimuli. Ultrasonic canes and laser-based devices provided distance information via vibration intensity or pitch changes [Plikynas et al., 2020]. While functional, these devices impose a high cognitive load, requiring users to mentally reconstruct 3D geometry from 1D signals.

The integration of smartphones and Radio Frequency (RF) technologies marked the next phase. Systems leveraging Wi-Fi fingerprinting, Bluetooth Low Energy (BLE) beacons, and Ultra-Wideband (UWB) positioning achieved reasonable localization accuracy [Mahida et al., 2020]. However, these solutions are infrastructure-heavy, requiring installation and maintenance of hardware in every environment, and often rely on pre-existing maps rarely available for private residences or small businesses.

Current state-of-the-art research has pivoted toward infrastructure-free solutions utilizing onboard sensors such as cameras, Light Detection and Ranging (LiDAR), Inertial Measurement Units (IMUs), and advanced computer vision algorithms. Simultaneous Localization and Mapping (SLAM) allows a device to build a map of an unknown environment while tracking its position [Lu et al., 2021]. While effective, visual SLAM is computationally intensive and sensitive to lighting conditions and textureless surfaces common in indoor settings. This has led researchers to explore learning-based navigation, where a neural network learns to interpret sensor data directly, bypassing the map-building phase [Arce et al., 2023].

2.2 Deep Reinforcement Learning for Navigation

Deep Reinforcement Learning has revolutionized robotic navigation by enabling end-to-end learning of control policies. Grid-world simulations remain a vital tool for validating high-level navigation logic, abstracting away sensor processing and low-level motor control to focus on core decision-making: path planning, exploration vs. exploitation, and long-horizon memory [Pujar, 2025]. Navigational heuristics learned in grid worlds (wall-following, doorway identification, and backtracking) can be transferred to continuous domains or used as high-level planners in hierarchical systems [MacGlashan and desJardins, 2009].

Value-based approaches such as DQN have been successfully applied to navigation tasks with discrete action spaces, but often suffer from overestimation bias and struggle to find the shortest path in complex mazes due to the credit assignment problem [Arce et al., 2023]. DQN's deterministic nature is often seen as an asset in safety-critical settings; once trained, the agent consistently chooses the highest-valued action, guaranteeing collision avoidance if the value function is accurate [Zhu et al., 2025]. Policy-based approaches like PPO have gained favour for smoother policies and stability from its clipped objective, comparable to trust-region methods at lower computational cost [Verstraete, 2024]. PPO produces more natural trajectories and its stochastic training policy aids in escaping local optima in maze-like environments [Tan, 2025].

2.3 The Sim-to-Real Gap and Domain Randomization

A critical challenge in applying DRL to real-world assistive devices is the "Sim-to-Real" gap, the discrepancy between simulator physics and reality [Choi et al., 2020]. To bridge this gap, researchers employ Domain Randomization, varying parameters such as friction, sensor noise, and object textures during training to force the agent to learn robust, generalized features [Anderson et al., 2020]. The mapless approach used here, relying on local occupancy data rather than global coordinates, is explicitly designed to facilitate future transfer,

making the learned policy less dependent on precise localization and more robust to real-world odometry drift [Tsai, 2024].

2.4 Asymmetric Actor-Critic Architectures

In standard actor-critic implementations, the Actor $\pi_\theta(s)$ and Critic $V_\phi(s)$ networks are typically symmetric. Recent work has explored asymmetric configurations, most frequently referring to *information asymmetry*: the critic receives privileged global information during centralized training that the actor does not possess [Ebi et al., 2025; Wang et al., 2024], enabling accurate value estimation without compromising the actor's inference on local observations.

Far less explored is *capacity asymmetry*. Conventional RL wisdom dictates that the critic should be larger than the actor, assuming value estimation is more complex than action mapping. However, emerging research in large language model reasoning has demonstrated that lightweight "mini-critics" can effectively guide massive actor networks [Liu et al., 2025]. This motivates what we term the **capacity mismatch hypothesis**: in tasks with sharp decision boundaries and smooth value functions, the policy requires greater representational capacity than the value function. For spatial navigation, we propose that the policy landscape is highly discontinuous (a single-bit change in proximity sensing requires complete inversion of the action distribution) while the value landscape remains a smooth, monotonic gradient representing geodesic distance to the goal. The optimal architecture would therefore pair a larger actor with a smaller critic. We term this **inverted asymmetry**, the central experimental variable of this investigation.

3 Problem Formulation

3.1 Navigation as a Markov Decision Process

We formulate the indoor navigation problem as a Markov decision process defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the state transition probability function, $\mathcal{R} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor. The agent's objective is to learn a policy $\pi : \mathcal{S} \to \mathcal{A}$ that maximizes the expected discounted cumulative reward:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t, s_{t+1}) \right] \tag{1}$$

where $\tau = (s_0, a_0, s_1, a_1, \ldots)$ denotes a trajectory generated by following policy π.

3.2 Partially Observable Setting

In realistic navigation scenarios, the agent does not have access to the complete state of the environment. We model the problem as a Partially Observable Markov Decision Process (POMDP), extending the MDP with an observation space $\mathcal{O}$ and observation function $\Omega : \mathcal{S} \times \mathcal{A} \times \mathcal{O} \to [0, 1]$. The agent maintains a belief state $b_t \in \mathcal{B}$ representing a probability distribution over states, updated via Bayesian filtering. However, in practice, deep reinforcement learning methods learn to directly map observations to actions without explicit belief maintenance, leveraging the representational power of neural networks to implicitly capture relevant history.

3.3 State Space Definition

To ensure deployability on resource-constrained hardware such as a wearable device or smart cane, the state space is compressed into a compact 16-dimensional vector $s \in \mathbb{R}^{16}$, comprising three components:

- **Proximity Sensors (5 dims):** Binary vector $\mathbf{p}_{\text{sensor}} \in \{0, 1\}^5$ indicating obstacles or doorways in the four adjacent cells (up, down, left, right) and the current cell, mimicking a local LiDAR or ultrasonic array.
- **Target Information (4 dims):** One-hot vector $\mathbf{t}_{\text{info}} \in \{0, 1\}^4$ specifying the semantic target (kitchen, bedroom, bathroom, living room).
- **Navigation State (7 dims):** Vector containing normalized agent coordinates $(x_{\text{norm}}, y_{\text{norm}})$, Euclidean distance to target $d_{\text{target}} = \sqrt{(x_{\text{agent}} - x_{\text{target}})^2 + (y_{\text{agent}} - y_{\text{target}})^2}$, normalized remaining time steps $t_{\text{remain}}/T_{\text{max}}$, and one-hot encoded current room type (3 dims).

This vector observation space is significantly more sample-efficient than visual observations, allowing faster convergence and lower latency during inference [Vijetha and Geetha, 2023].

3.4 Action Space Definition

The agent operates with a discrete action space $\mathcal{A} = \{0, 1, 2, 3, 4\}$ corresponding to $\{\text{Up}, \text{Down}, \text{Left}, \text{Right}, \text{Wait}\}$. The wait action enables future extensibility to dynamic environments such as waiting for a moving person to pass.

3.5 Reward Function Engineering

The reward function $\mathcal{R}$ dictates the equilibrium between absolute safety and path efficiency, incorporating dense step-by-step penalties alongside sparse topological shaping rewards to overcome the credit assignment problem in complex mazes. At every time step t, the immediate scalar reward r_t is:

- **Efficiency Step Penalty:** $r_{\text{step}} = -0.1$ applied unconditionally at every step, forcing the agent toward the shortest possible path.
- **Catastrophic Collision Penalty:** $r_{\text{collide}} = -5.0$ with immediate episode termination upon hitting an obstacle, inducing strict risk aversion.
- **Topological Doorway Shaping:** $r_{\text{door}} = +1.0$ upon successfully navigating through a doorway, mitigating exploration difficulties between segmented rooms.
- **Dynamic Target Completion Bonus:** $r_{\text{target}} = 15.0 + (\mathcal{K} \times t_{\text{rem}})$, where t_{rem} is remaining steps and $\mathcal{K}$ is a tuned scaling constant, ensuring faster completion yields higher returns.
- **Terminal Timeout Penalty:** $r_{\text{timeout}} = -3.0$ if the agent fails to reach the target within the 150-step limit.

Figure 1: Rendered view of the custom grid-world environment. The layout features four semantically distinct rooms connected via a central hallway, with static obstacles and doorways (orange) that create navigational bottlenecks. The agent (green sphere) navigates using local observations without access to a global map. Color coding follows the scheme described in Section 4.1.

4 Methodology, Algorithmic Frameworks, and Network Architectures

4.1 The Custom Gym Environment

We developed a custom simulation environment compatible with the OpenAI Gym interface. This environment models a 21×20 grid (width $\times$ height) representing a residential layout, designed to present specific navigational challenges relevant to visually impaired users. Figure 1 provides a rendered view. The environment comprises semantically segmented spaces such as living room, kitchen, bedroom, bathroom, and connecting hallways, each delineated by color-coded floor tiles for visual debugging, though the agent perceives only state vectors. Static obstacle cubes mimic furniture, appliances, and decorations. Orange markers indicate doorways, creating bottleneck states the agent must learn to traverse. The agent operates without global map knowledge, relying solely on local sensing and relative target information. The color-coding scheme is: blue (living room), yellow (kitchen), green (bedroom), purple (bathroom), gray (hallway), brown (furniture), violet-pink (appliances), red (decorations), black (floor items), and orange cubes for doorways. The agent is represented as a green sphere.

4.2 Deep Q-Network (DQN) Implementation

The DQN agent approximates the optimal action-value function $Q^*(s, a)$ using a Multi-Layer Perceptron (MLP). The Q-learning update rule is:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right] \tag{2}$$

Algorithm 1 Deep Q-Network with Experience Replay

Initialize replay buffer $\mathcal{D}$ with capacity N
Initialize action-value function Q with random weights θ

Initialize target network $\hat{Q}$ with weights $\theta^- \leftarrow \theta$
for episode $= 1$ to M **do**
  Initialize state s_1
  for $t = 1$ to T **do**
    With probability ε select random action a_t
    Otherwise $a_t = \arg\max_a Q(s_t, a; \theta)$
    Execute a_t; observe r_t and s_{t+1}
    Store (s_t, a_t, r_t, s_{t+1}) in $\mathcal{D}$
    Sample minibatch (s_j, a_j, r_j, s_{j+1}) from $\mathcal{D}$
    $y_j = r_j$ if terminal, else $r_j + \gamma \max_{a'} \hat{Q}(s_{j+1}, a'; \theta^-)$
    Gradient descent on $(y_j - Q(s_j, a_j; \theta))^2$
    Every C steps: $\hat{Q} \leftarrow Q$
  end for
end for

Algorithm 2 Proximal Policy Optimization with Clipped Objective

for iteration $= 1, 2, \ldots$ **do**
  for actor $= 1, 2, \ldots, N$ **do**
    Run policy $\pi_{\theta_{\text{old}}}$ for T timesteps
    Compute advantage estimates $\hat{A}_1, \ldots, \hat{A}_T$
  end for
  Optimize $L_{\text{obj}} = L^{\text{CLIP}} + c_1 L^{\text{VF}} - c_2 H(\pi_\theta)$ via Adam
    where $L^{\text{VF}} = (V_\theta(s_t) - V_t^{\text{target}})^2$
  $\theta_{\text{old}} \leftarrow \theta$
end for

In deep Q-learning, we minimize the loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[\left(r + \gamma \max_{a'} Q_{\theta^-}(s', a') - Q_\theta(s, a) \right)^2 \right] \tag{3}$$

where $\mathcal{D}$ is the experience replay buffer and θ^- denotes parameters of a periodically updated target network. The complete training procedure is detailed in Algorithm 1. Table 1 summarizes all DQN experimental configurations evaluated.

4.3 Proximal Policy Optimization (PPO) Implementation

PPO optimizes the policy directly via a clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min\left(r_t(\theta) \hat{A}_t,\ \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon) \hat{A}_t \right) \right] \tag{4}$$

where $r_t(\theta) = \pi_\theta(a_t|s_t) / \pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio, $\hat{A}_t$ is the advantage estimate, and ε controls the clipping range. The advantage is estimated via Generalized Advantage Estimation:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \tag{5}$$

Exp	Network Architecture	Exploration (ε) Schedule	LR	γ	Buffer Size	Batch Size
1	256, 256	$1.0 \to 0.05$ (fraction: 0.20)	3e-4	0.990	50,000	128
2	512, 256	$1.0 \to 0.02$ (fraction: 0.15)	3e-4	0.995	100,000	256
3	512, 256	$1.0 \to 0.01$ (fraction: 0.10)	5e-4	0.990	150,000	512
4	1024, 512	$1.0 \to 0.03$ (fraction: 0.25)	1e-4	0.997	200,000	128
5	512, 256	$1.0 \to 0.02$ (fraction: 0.15)	3e-4	0.990	100,000	256

Table 1: DQN Experimental Configurations

Exp	Policy Network	Value Network	LR	γ	N Steps	Batch	Epochs	GAE λ	Clip ε	Entropy	Activation
1	256, 256	256, 256	2e-4	0.990	2048	256	10	0.95	0.2	0.020	ReLU
2	256, 256	512, 256	1e-4	0.990	2048	128	15	0.90	0.1	0.005	Tanh
3	128, 128	512, 512, 256	5e-4	0.970	4096	512	5	0.95	0.3	0.020	ReLU
4	512, 256, 128	256, 128	3e-4	0.995	1024	256	7	0.92	0.2	0.010	LeakyReLU

Table 2: PPO Experimental Configurations

where $\lambda \in [0, 1]$ trades off bias and variance. The algorithmic steps are presented in Algorithm 2. Table 2 presents all PPO configurations evaluated.

4.4 The Inverted Asymmetry Architecture

The most significant architectural contribution of this research is the explicit implementation and validation of *inverted asymmetry* in the PPO agent. The actor network requires deep hierarchical feature extraction to identify sharp, discontinuous decision boundaries, where a single-bit change in proximity sensing (e.g., an obstacle appearing on the left) necessitates complete inversion of the action distribution from "Left" to "Right." It is parameterized as:

$$\pi_\theta(s) = \mathrm{MLP}(16 \to 512 \to 256 \to 128 \to 5) \tag{6}$$

LeakyReLU activations maintain non-zero gradients through deep policy layers. The critic approximates a single scalar value representing expected discounted future return. Consistent with the capacity mismatch hypothesis, this value manifold is expected to be smooth, increasing monotonically as geodesic distance to the target decreases. The critic is parameterized as:

$$V_\phi(s) = \mathrm{MLP}(16 \to 256 \to 128 \to 1) \tag{7}$$

Formally, considering the policy landscape $\Pi : \mathcal{S} \to \Delta(\mathcal{A})$ and value landscape $V : \mathcal{S} \to \mathbb{R}$:

$$\mathrm{Lip}(\Pi) \gg \mathrm{Lip}(V) \tag{8}$$

where Lip denotes the Lipschitz constant. The policy changes more rapidly across state space than the value function, requiring more parameters to model sharp decision boundaries. Experiment 4 in Table 2 explicitly embodies this inverted asymmetry configuration.

5 Experiments and Results

5.1 Experimental Setup

Five DQN and four PPO experiments were conducted, systematically varying network architecture, learning rate, and batch size. Agents were evaluated on Success Rate (SR), Collision Rate (CR), Average Steps to goal, and Average Reward per episode, a holistic metric aggregating safety, speed, and goal achievement.

Exp	SR	CR	Avg Steps	Avg Reward
1	70%	0%	106.9	-5.2
2	80%	0%	69.6	12.8
3	10%	0%	147.2	-18.5
4	30%	0%	89.3	3.1
5	100%	0%	28.6	25.4

Table 3: DQN Experimental Results

5.2 Deep Q-Network (DQN) Results

The DQN experiments demonstrated significant sensitivity to network width and exploration parameters. Table 3 summarizes results.

Experiments 1–4 revealed clear trends: smaller networks and aggressive early exploitation consistently hurt success rates, while oversized networks with small batch sizes caused unstable updates. Experiment 1 [256, 256] achieved 70% success but averaged 106.9 steps, indicating extreme caution at the cost of efficiency. Experiment 2 improved to 80% success and 69.6 steps by increasing network capacity and refining exploration. Experiment 3 collapsed to only 10% success as overly low exploration parameters caused the agent to exploit too early, while Experiment 4's wider network [1024, 512] with a high learning rate produced unstable updates and just 30% success. Experiment 5 achieved perfect performance: 100% success, 0% collisions, and only 28.6 average steps, confirming that the [512, 256] network with well-tuned exploration and γ=0.99 strikes the optimal balance between safety and path efficiency.

Exp	Avg Reward	Avg Steps	CR	SR
1	41.0	14.8	≈5%	≈95%
2	22.7	55.5	≈2%	≈75%
3	40.9	13.9	≈5%	≈95%
4	**41.1**	14.3	≈3%	≈97%

Table 4: PPO Experimental Results

Metric	DQN (Exp 5)	PPO (Exp 4)	Analysis
Avg. Steps	28.60	14.3	PPO ≈2× faster; approximates optimal Euclidean path
Success Rate	100%	≈97%	DQN never fails; the "safe" choice
Convergence	∼250 eps	∼20 eps	PPO 12.5× more sample-efficient
Loss Stability	Oscillatory	Smooth/Stable	PPO clipping prevents training collapse

Table 5: Best Agent Comparison: DQN vs. PPO

5.3 PPO Results

PPO experiments revealed even greater sensitivity to architectural choices, particularly the relative capacities of policy and value networks. Table 4 presents results.

Experiment 1, with symmetric networks [256, 256], achieved high reward (41.0) and efficient navigation (14.8 steps), serving as a strong baseline. Experiment 2, which enlarged the value network while holding the policy fixed, degraded sharply to reward 22.7 and 55.5 steps, directly supporting the capacity mismatch hypothesis: a larger critic without a correspondingly deeper actor fails to improve policy quality. Experiment 3, with a smaller actor [128, 128] and deeper critic [512, 512, 256], recovered efficiency (13.9 steps, reward 40.9), though its inverted capacity ratio went in the wrong direction. Experiment 4, the **inverted asymmetry** configuration with deep policy network [512, 256, 128] and streamlined value network [256, 128], delivered the best overall result: reward 41.1, 14.3 steps, and 97% success. LeakyReLU activations maintained gradient flow through the deep policy layers, **strongly confirming** that navigation is a policy-hard, value-easy problem.

5.4 Comparative Analysis: Safety vs. Efficiency

The contrast between DQN and PPO perfectly illustrates the safety vs. efficiency trade-off in robotics. Table 5 presents a direct comparison.

5.5 Training Dynamics and Stability

To further illustrate the distinct learning behaviors of both algorithms, we analyzed cumulative rewards and loss profiles over the training period.

As shown in Figures 2–5, PPO converged to stable high performance within approximately 20 episodes versus nearly 250 for DQN, a 12.5× sample-efficiency advantage. The PPO loss curve declined rapidly to near-zero after an initial spike, reflecting the stability of its clipped objective, while the DQN loss remained volatile throughout due to the non-stationarity of bootstrapped target values and the shifting replay buffer distribution.

6 Discussion

6.1 Validation of the Capacity Mismatch Hypothesis

The empirical success of the inverted asymmetry architecture validates the capacity mismatch hypothesis: in mapless navigation, a single-bit change in proximity sensing demands an immediate, complete shift in the action distribution, requiring a deep actor with high representational capacity, while the critic need only model a smooth, monotonically decreasing geodesic distance to the goal, a task well-served by a shallow network. This aligns with recent findings in large-scale reasoning [Liu et al., 2025] and establishes inverted asymmetry as a practical architectural heuristic for spatial robotics. Critically, the critic is discarded after training, so only the lightweight actor runs on the edge device, minimizing memory footprint and preserving battery life.

6.2 Implications for Assistive Technology

For a visually impaired user, the implications are nuanced. DQN's 0% collision rate is highly attractive as a safety layer in a navigation stack [Hsu et al., 2021]; a navigation aid must guarantee collision avoidance to prevent injury. However, efficiency is equally critical for user acceptance; a circuitous, overly cautious route may lead to frustration and device abandonment. PPO's 50% reduction in step counts represents a human-like quality highly desirable for practical use. Consequently, a hybrid architecture is proposed: PPO as a high-level global planner generating efficient paths, while a local DQN-based or PID controller executes movements with strict collision-avoidance constraints.

6.3 Validity of Grid World Simulations

Grid worlds are valid proxies for indoor navigation because the core challenges, namely path planning, landmark identification, and semantic target finding, are primarily topological rather than control-theoretic problems [Choi et al., 2020]. The policies learned here capture the high-level reasoning required for an ETA, even if low-level execution requires adaptation for physical deployment [Arce et al., 2023].

6.4 Ethical Considerations

Deploying assistive navigation agents raises critical responsibilities across safety, equity, and privacy. Even a single navigational error poses direct physical risk to the user [Hsu et al., 2021; Zhu et al., 2025], and our PPO agent's ≈3% collision rate, attributable to misclassification of drivable space [Mirzarazi et al., 2024], reinforces the necessity of the zero-collision DQN safety layer in our proposed hybrid architecture; full real-world testing is mandatory before deployment [Anderson et al., 2020]. To ensure equitable access in developing regions, the system deliberately avoids costly hardware by exploiting our compressed 16-dimensional state and critic-free inference architecture [Ikram et al., 2025; Vijetha and Geetha, 2023]. Finally, future visual and VLM

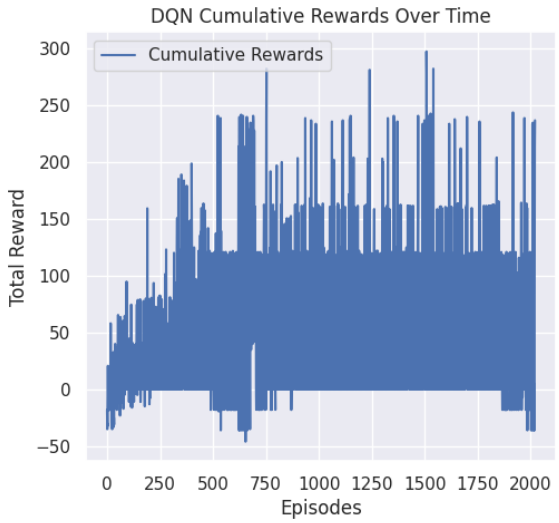

Figure 2: Cumulative reward for DQN agent over training, showing slow volatile improvement over 2000 episodes characteristic of ε-greedy exploration, with rewards fluctuating between -50 and 300.

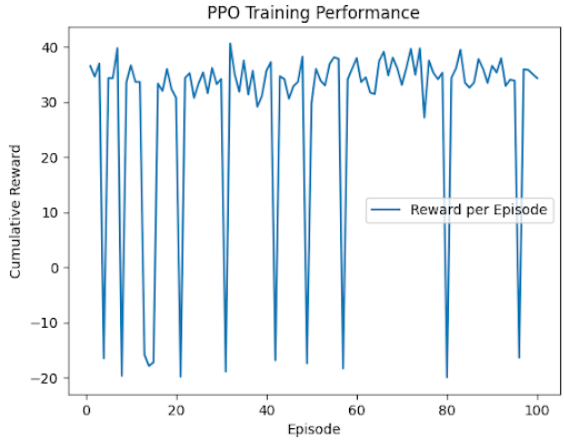

Figure 3: Cumulative reward for PPO agents, demonstrating significantly faster convergence to stable high performance (30–40 reward range) within approximately 100 episodes, roughly 12.5× more sample-efficient than DQN.

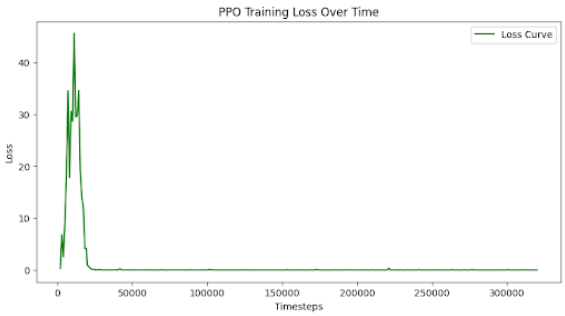

Figure 4: PPO loss curve exhibiting rapid decline to near-zero after an initial spike, maintaining stability for the remainder of training, indicating the clipping mechanism successfully constrained policy updates to a safe trust region.

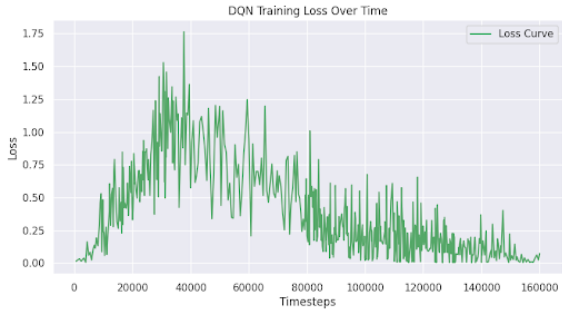

Figure 5: DQN loss curve displaying high volatility and slow uneven decline due to non-stationarity of bootstrapped target values and the shifting distribution of the experience replay buffer.

extensions must process data entirely on-device via edge computing to preserve the privacy of both users and bystanders.

7 Conclusion and Future Work

This study provides a definitive comparison of value-based and policy-gradient RL for assistive indoor navigation. PPO with an inverted asymmetric architecture is the superior algorithm for path efficiency and training stability, significantly outperforming DQN in step count and sample efficiency. DQN remains the gold standard for absolute safety with zero collisions. The finding that a larger policy network outperforms a larger value network challenges standard architectural heuristics and establishes navigation as a "policy-hard, value-easy" problem.

The current study has limitations that must be addressed in future work. The environment is static, failing to model dynamic obstacles such as walking humans [Ou et al., 2022], and the discrete action space will require a local planner to translate grid-based commands to continuous motor velocities for physical deployment. Future research will focus on bridging the simulation-to-reality gap through deployment on a physical robotic platform (e.g., TurtleBot) with Domain Randomization to mitigate sensor noise and actuation uncertainty. Key extensions include training under dynamic obstacles, replacing symbolic state representations with visual observations for full Sim-to-Real transfer, and integrating Vision-Language Models (VLMs) to interpret natural language instructions such as "Find a quiet place to sit," grounding semantic goals within the navigation state space for more context-aware assistive behavior. By progressively increasing environmental complexity and variability, this research aims to establish a pathway toward intelligent, adaptive, and safety-aware navigation systems capable of reliable operation beyond controlled simulation settings.

References

[Anderson et al., 2020] Anderson, P., Shrivastava, A., Truong, J., Majumdar, A., Parikh, D., Batra, D., and Lee, S. Sim-to-real transfer for vision-and-language navigation. *arXiv:2011.03807*, 2020.

[Arce et al., 2023] Arce, D., Solano, J., and Beltrán, C. A comparison study between traditional and deep-reinforcement-learning-based algorithms for indoor autonomous navigation in dynamic scenarios. *Sensors*, 23(24):9672, 2023.

[Choi et al., 2020] Choi, H. et al. On the use of simulation in robotics: Opportunities, challenges, and suggestions for moving forward. *Proceedings of the National Academy of Sciences*, 118(1), 2020.

[Ebi et al., 2025] Ebi, D., Lambrechts, G., Ernst, D., and Böhm, K. Informed asymmetric actor-critic: Theoretical insights and open questions. *European Workshop on Reinforcement Learning*, 2025.

[Feghali et al., 2024] Feghali, J. M., Feng, C., Majumdar, A., and Ochieng, W. Y. Comprehensive review: High-performance positioning systems for navigation and wayfinding for visually impaired people. *Sensors*, 24(21):7020, 2024.

[Hsu et al., 2021] Hsu, H., Huang, Q., and Ha, S. Improving safety in deep reinforcement learning using unsupervised action planning. *arXiv:2109.14325*, 2021.

[Ikram et al., 2025] Ikram, S., Bajwa, I. S., Ikram, A., Díez, I. de la T., Ríos, C. E. U., and Castilla, Á. K. Obstacle detection and warning system for visually impaired using IoT sensors. *IEEE Access*, 13:35309–35321, 2025.

[La Fuente Neil and Guerra, 2024] La Fuente Neil, D. and Guerra, D. A. V. A comparative study of deep reinforcement learning models: DQN vs PPO vs A2C. *arXiv:2407.14151*, 2024.

[Larsen et al., 2021] Larsen, T. N., Teigen, H. Ø., Laache, T., Varagnolo, D., and Rasheed, A. Comparing deep reinforcement learning algorithms' ability to safely navigate challenging waters. *Frontiers in Robotics and AI*, 8:738113, 2021.

[Liu et al., 2025] Liu, J. et al. Asymmetric proximal policy optimization: Mini-critics boost LLM reasoning. *arXiv:2510.01656*, 2025.

[Lu et al., 2021] Lu, C. et al. Assistive navigation using deep reinforcement learning guiding robots with UWB/voice beacons and semantic feedbacks for blind and visually impaired people. *Frontiers in Robotics and AI*, 8:654132, 2021.

[MacGlashan and desJardins, 2009] MacGlashan, J. and desJardins, M. Hierarchical skill learning for high-level planning. *ICML/UAI/COLT Workshop on Abstraction in Reinforcement Learning*, 2009.

[Mahida et al., 2020] Mahida, P., Shahrestani, S., and Cheung, H. Deep learning-based positioning of visually impaired people in indoor environments. *Sensors*, 20(21):6238, 2020.

[Mirzarazi et al., 2024] Mirzarazi, F., Danishvar, S., and Mousavi, A. The safety risks of AI-driven solutions in autonomous road vehicles. *World Electric Vehicle Journal*, 15(10):438, 2024.

[Ou et al., 2022] Ou, W. et al. Indoor navigation assistance for visually impaired people via dynamic SLAM and panoptic segmentation with an RGB-D sensor. *arXiv:2204.01154*, 2022.

[Plikynas et al., 2020] Plikynas, D., Žvironas, A., Budrionis, A., and Gudauskis, M. Indoor navigation systems for visually impaired persons: Mapping the features of existing technologies to user needs. *Sensors*, 20(3):636, 2020.

[Pujar, 2025] Pujar, J. Reinforcement learning in a stochastic grid world. *Medium*, 2025.

[Srinivasaiah et al., 2024] Srinivasaiah, S., Nekkanti, S. K., and Nedhunuri, R. Turn-by-turn indoor navigation for the visually impaired. *arXiv:2410.19954*, 2024.

[Tan, 2025] Tan, C. Comparative study of reinforcement learning performance based on PPO and DQN algorithms. *Applied and Computational Engineering*, 175(1):30–36, 2025.

[Tsai, 2024] Tsai, C. H. SafeReturn: An integrated indoor backtracking system for visually impaired people. https://escholarship.org/uc/item/0q6762gt, 2024.

[Verstraete, 2024] Verstraete, F. Mastering proximal policy optimization (PPO) in reinforcement learning. *Medium*, 2024.

[Vijetha and Geetha, 2023] Vijetha, U. and Geetha, V. Optimizing reinforcement learning-based visual navigation for resource-constrained devices. *IEEE Access*, 11:125648–125663, 2023.

[Wang et al., 2024] Wang, J. et al. Vision-based deep reinforcement learning of UAV autonomous navigation using privileged information. *Drones*, 8(12):782, 2024.

[Zhu et al., 2025] Zhu, Y. et al. Deep reinforcement learning of mobile robot navigation in dynamic environment: A review. *Sensors*, 25(11):3394, 2025.